

Laws and chance

PHIL UA-78 Metaphysics

Prof. Cian Dorr

9th September 2026

The most important modality for philosophy is metaphysical modality.

The *second* most important modality for philosophy—and arguably the most important one for science—is *nomic* modality, the notion of necessity and possibility associated with *laws of nature*. ('Nomic' comes from the Greek *nomos*, law.)

We have good reason to think that it's nomically necessary that every particle subject to [almost] no external forces moves in a [nearly] straight line. But it seems *metaphysically* possible for such a particle not to move in a straight line—the supposition doesn't seem to “contain a contradiction”.

Laws of nature and nomic necessities

- ▶ Some philosophers use 'law of nature' as synonymous with 'nomically necessary state of affairs'.
- ▶ Others use 'law of nature' in a more demanding way. On this use, even if it's a law of nature that no particle travels faster than light, and therefore nomically necessary that no *particle that was once part of a banana* travels faster than light, maybe the latter is not a law of nature.

It's uncontroversial that all laws of nature are nomically necessary.

It's *pretty* uncontroversial that everything nomically necessary follows with metaphysical necessity from the laws of nature.

If you distinguish the two notions, you might try to define one in terms of the other.

- ▶ 'Nomically necessary state of affairs' might mean 'State of affairs that follows with metaphysical necessity from the totality of all laws of nature'.
- ▶ Or, 'Law of nature' might mean something like 'Member of a simple set of axioms from which all nomically necessary truths follow with metaphysical necessity'.

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We will be focusing on the notion of nomic necessity, which seems to me to be more important. Questions about which nomic necessities are laws and which are not do not seem to play a big role in science or philosophy.

The view that nomic and metaphysical modality are identical

There are some philosophers (most influentially, Sydney Shoemaker) who maintain that nomic modality is identical to metaphysical modality.

Their picture is that just as containing hydrogen is part of what it is to be water, *moving in a straight line* is part of what it is to be a particle not subject to any forces. . . .

- ▶ Hardest cases: *Spacetime is four-dimensional, Elementary particles come in at most three families,*

I'll often talk in ways that take it for granted that this view is false.

Historical modality and determinism

There's another kind of modality that's closely connected to nomic modality.

When I say 'Since the particle was over here and not subject to any forces, it was impossible for it to get all the way to the target', I don't mean that it was nomically impossible—the particle could have started out near the target!

Rather, what I plausibly mean is that it is nomically impossible that *the particle makes it to the target and the course of history up to some salient time is just as it actually is*.

- Say that it is **historically possible that p at t** iff the complete truth h about the course of history up to t is such that it's nomically possible that $(p$ and $h)$.

The complete truth about the course of history up to t

In interpreting this definition, we need to be careful about what counts as 'The complete truth about the course of history up to t '.

- We don't want to reason like this: 'One property of World War I is that it was first of two world wars to break out in a 30-year period. So there's a fact about the course of history before 1920—namely that it included a war with this property beginning in 1914—which nomically (and indeed metaphysically) necessitates that another world war would break out by 1944. So, it was historically impossible in 1920 for another world war not to break out before 1944.'

When we say 'facts about the course of history up to t ', we mean facts about the **intrinsic state** of the world during that period.

Determinism

We can define **determinism** as the view that *every truth is historically necessary at every time*.

Cashing out the definition: every truth is nomically necessitated by the facts about the course of history up to any given time.

Or in terms of 'law of nature': every truth is *metaphysically* necessitated by the facts about the course of history up to any given time together with the laws of nature.

Determinism is widely regarded as a live option in physics. Before the discovery of quantum mechanics, most physicists regarded it as a settled matter. Its status since then is more debatable: there are competing 'interpretations' of QM, some of which preserve determinism and others of which do not.

Some philosophers—**free will incompatibilists**—think that no-one could have *free will* in a world where determinism is true. Others—**free will compatibilists**—think that it's possible for people to have free will even in such a world.

Among the free will incompatibilists, *some* think that since we know we do have free will, we can rule out determinism. Others think that we should not be confident that determinism is false, and therefore should not be confident that we have free will. Some even think that we should be confident that we *don't* have free will, and some think that free will is metaphysically impossible!

We'll be talking about this more in 2 weeks.

Chance

Chance

The notion of nomic necessity is closely connected to another notion, that of *objective chance*.

‘Chance’ and ‘probability’ are used in different ways.

- ▶ Sometimes it is ‘epistemic’, as in ‘Since I have searched my house for an hour and not found my keys, there’s a high chance that I left them somewhere outside my house’. This sort of use has to do with *how confident* some relevant people are, or should be, that things are a certain way.
- ▶ *Objective* chance is what we would be talking about if we said things like ‘Since this is a fair coin, the chance that it’ll land heads the next time it is tossed is $1/2$ ’ or ‘The chance that your child will have blue eyes, given that one of you has blue eyes and the other has brown eyes and a parent with blue eyes, is $1/2$ ’.

Claims about objective chance don’t seem to be claims about us or how confident we should be that something is the case—they are about the world, and a major

The probability axioms

We can define a *probability function* to be a mapping π from states of affairs to real numbers such that:

1. $\pi(p \vee \neg p) = 1$ for any p .
2. $\pi(p) \leq \pi(q)$ whenever q follows from p (i.e., $p \rightarrow q$ is metaphysically necessary).
3. $\pi(p \vee q) = \pi(p) + \pi(q)$ whenever p and q are incompatible (i.e., $p \wedge q$ is metaphysically impossible).

Objective chances—and *conditional* objective chances—obey these axioms.

But so do lots of other things—e.g. it is widely held that an ideally rational person's degrees of confidence obey the axioms.

Often, when we talk about chance in an 'objective' way, we are talking about the chances at some particular time t .

These are generally understood in such a way that *facts about the course of history up to t* automatically have chance 1 at t .

However, there is some objective-chance-talk that isn't easily understood as having to do with chances at specific times.

Our two topics are closely connected, at least as follows: every nomically necessary truth has chance 1 at every time.

One possible view: *all and only* nomic necessities have chance 1 (timelessly); all and only historical necessities at t have chance 1 at t .

- ▶ This isn't quite right. It could be that when someone throws a dart at a dartboard, every individual point on the dartboard is such that its chance of not being hit is 1. But we certainly can't say that every point is such that it is historically impossible for it to be hit—the truth must be historically possible!
- ▶ One could get around this by invoking a *primitively conditional* notion of chance: p is nomically necessary just in case the chance of p given q is 1 whenever this is defined.

Doubts about the topic

Should we forget about lawhood?

Some philosophers have suggested that there is something mistaken or confused about the whole notion of a law of nature:

I think that our present use of the expression 'laws of nature' carries traces of the conception of Nature as subject to command. Whether these commands are conceived to be those of an impersonal deity or, as by the Greeks, of an impersonal fate, makes no difference here. The point, in either case, is that the sovereign is thought to be so powerful that its dictates are bound to be obeyed.

A.J. Ayer, 'What is a law of nature?'

Should we forget about chance?

By contrast, it is obviously not at all plausible that the notion of objective chance is some sort of theological holdover.

And while it's maybe not so hard to imagine scientists deciding to banish the notion of law of nature, it's very hard to imagine them deciding to banish talk of objective chance—so much of science seems to be primarily *about* trying to figure out the answers to questions about objective chance!!